

University of Dhaka



Assignment - 3

Course Title: Advanced Econometrics-II
Course Code: 408

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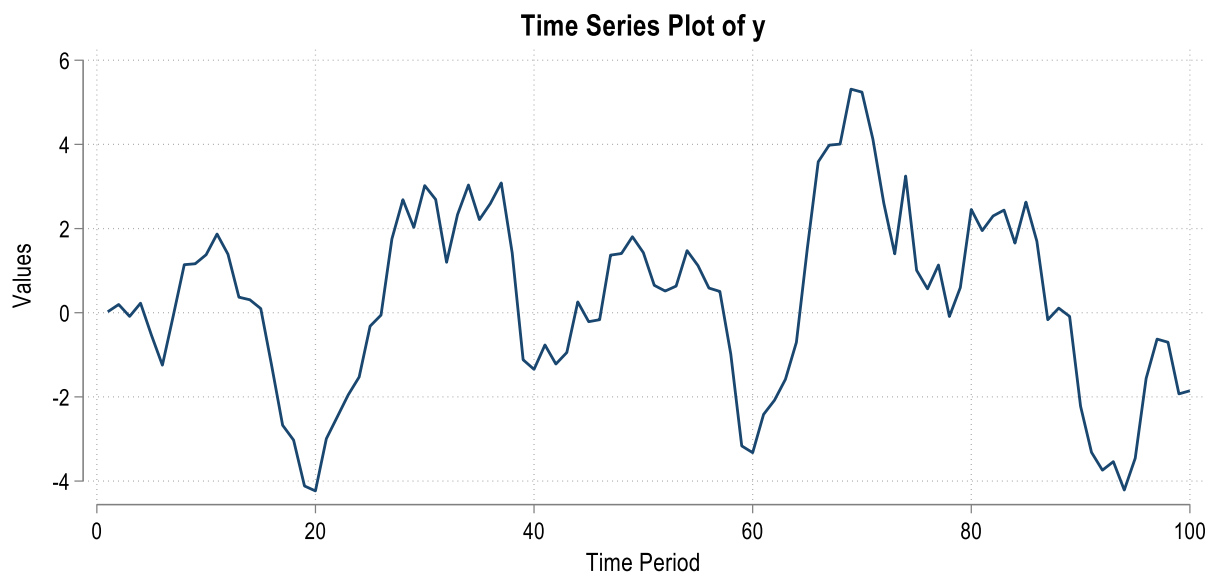
Question:

Attached is a STATA data file with a time series variable "y". First, using the Box-Jenkins methodology, determine which ARMA model this variable follows. After estimating the ARMA model, please forecast 10 steps ahead of "y" with a 95% confidence interval. Extra marks for better visual representation.

Solution:

Following the Box-Jenkins Methodology, we will follow (a) Identification (b) Estimation (c) Diagnostic Test, these 3 procedures to find the best Univariate ARMA model that this variable y follows.

(a) Identification:

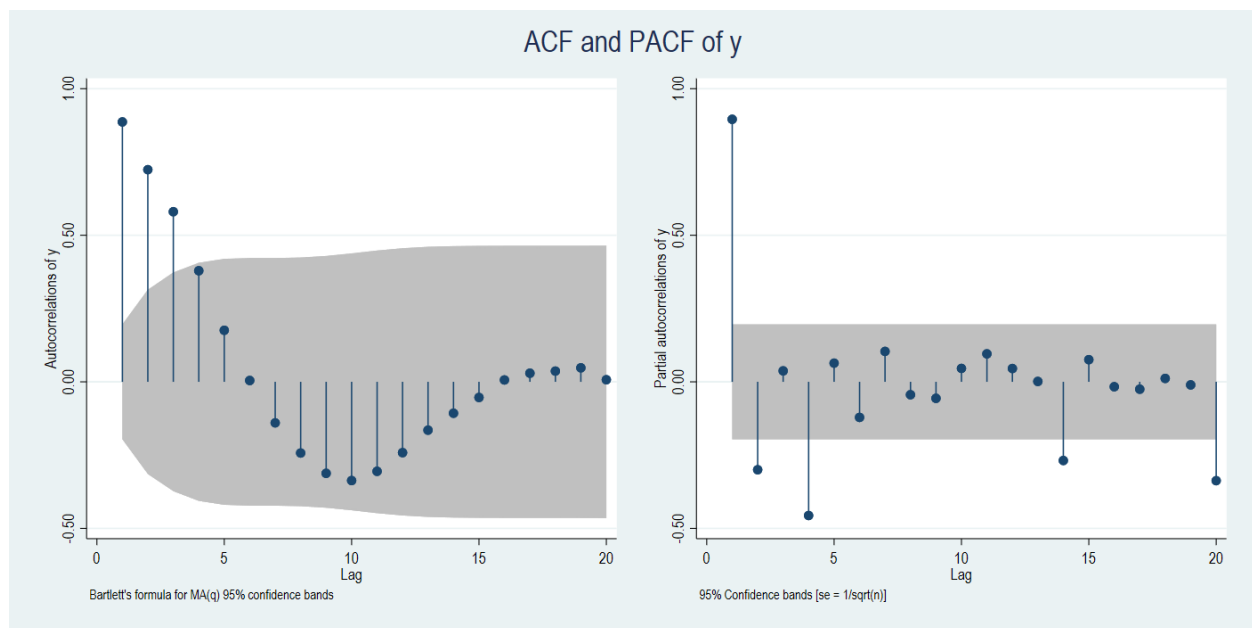


H0: Random walk without drift, $d = 0$

	Test	Dickey Fuller Critical Values		
	Statistics	1%	5%	10%
Z(t)	-3.622	-3.517	-2.894	-2.582

MacKinnon approximate p-value for Z(t) = 0.0054

The Time Series plot of the variable y clearly shows no significant trend. And the Augmented Dickey-Fuller Test also rejects the Null Hypothesis of a Unit Root Process at lag 4. So, we can conclude that variable y is stationary.



Both the Autocorrelation and Partial Auto-correlation Function of y variable show significant spikes till lag 3 and exponential decay in ACF. From Stata's varsoc command I found that the optimal AR lag is 4. So, I assume the presence of AR (4), ARMA (2,3), ARMA (4,1), ARMA (4,2), ARMA (4,3). Now I will estimate all the assumed models and compare their information criteria to get the parsimonious model.

Information Criteria:

<i>Model</i>	<i>N</i>	<i>LL (model)</i>	<i>df</i>	<i>AIC</i>	<i>BIC</i>
AR (4)	100	-124.669	7	261.338	276.969
ARMA (2,3)	100	-124.3935	7	262.7871	281.0233
ARMA (4,1)	100	-124.2639	7	262.5278	280.7639
ARMA (4,2)	100	-123.5619	8	263.1831	284.0245
ARMA (4,3)	100	-123.5017	9	265.0035	288.45

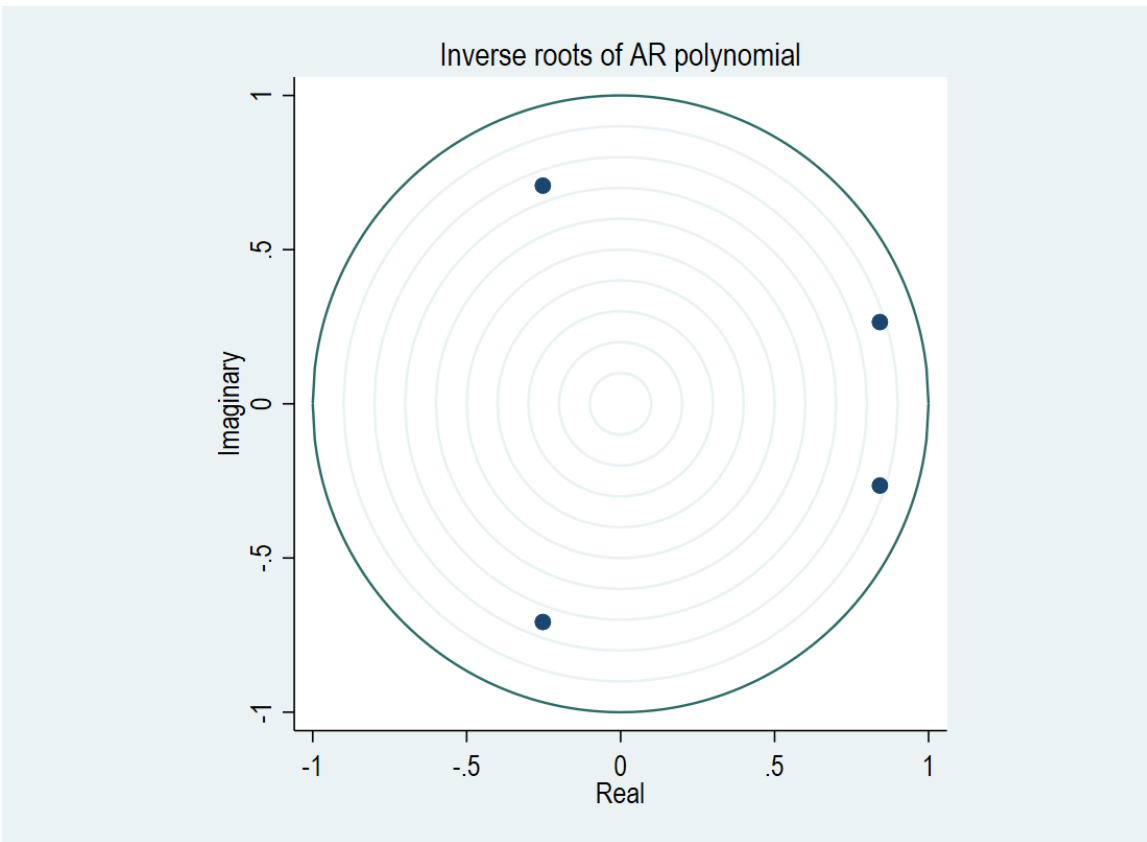
Among those guessed models, AR (4) yields the lowest AIC and BIC so this is my chosen Univariate model for the variable y.

(b) Estimation:

AR (4) / ARMA (4,0) regression:

y	Coef.	St.Err.	t-value	p-value	[95% Conf	Interval]	Sig
Constant	.235	.446	0.53	.598	-.639	1.108	
ar							
L	1.179	.085	13.89	0	1.012	1.345	***
L2	-.493	.122	-4.04	0	-.732	-.254	***
L3	.557	.159	3.50	0	.245	.868	***
L4	-.44	.09	-4.88	0	-.617	-.263	***
Constant	.831	.056	14.97	0	.722	.94	***
Mean dependent var		0.275	SD dependent var			2.162	
Number of obs		100	Chi-square			380.697	
Prob > chi2		.	Akaike crit. (AIC)			261.338	

*** $p < .01$, ** $p < .05$, * $p < .1$



After estimating the AR (4) model we can see all the lag coefficients of AR process are statistically significant at 1% confidence intervals. And the roots diagram shows us that all the roots of AR process are within the unit circle. So, we can say AR (4) or ARMA (4,0) model is acceptable for estimation.

(c) Diagnostic Test:

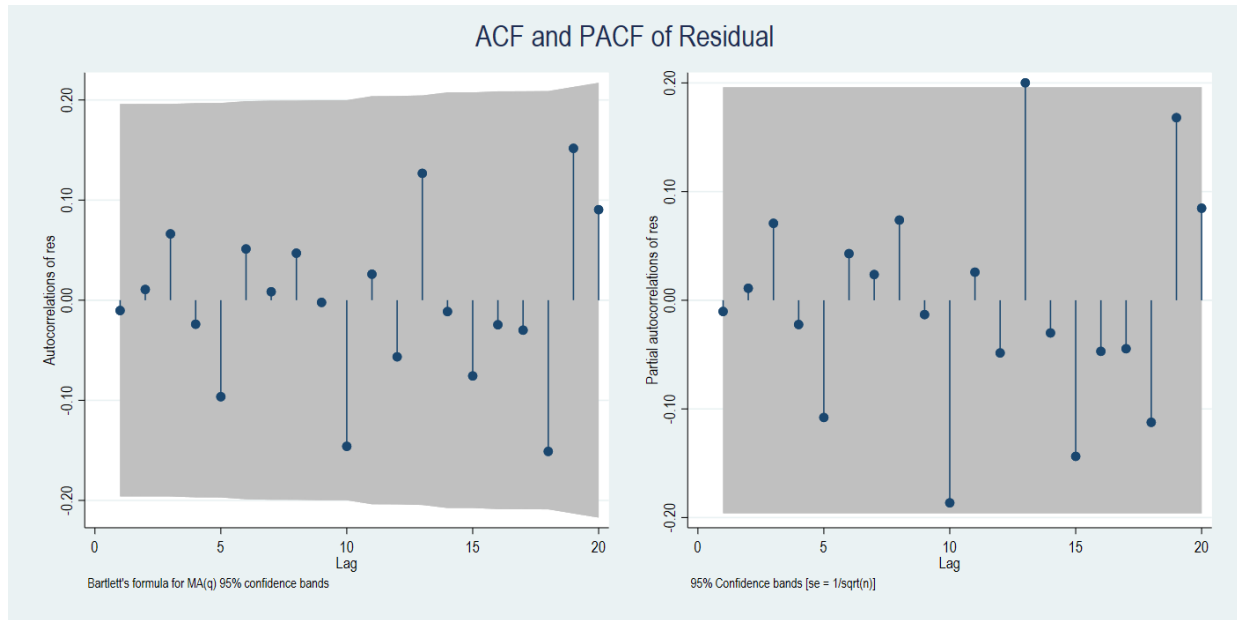
Portmanteau test for H_0 : white noise

Portmanteau (Q) statistics = 33.9823

Prob > chi2(40) = 0.7370

The Ljung-Box test or Q test at lag 40 do not reject the Null Hypothesis of the residuals being white noise. The Autocorrelation and Partial Autocorrelation function of the residuals also justify this white noise process for lag 20. This diagnostic test satisfies the residuals being

white noise and according to the steps of Box Jenkins Methodology AR (4) or ARMA (4,0) model is accepted as the univariate process of the variable y.



(d) 10 Steps Ahead Forecast:

